



12591 - MIRI and NIRCam imaging of star forming pillar

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Pillar-MIRI	MIRI Imaging	(1) Star-Forming-Pillar
	2	Pillar-NIRCam	NIRCam Imaging	(1) Star-Forming-Pillar
	3	Pillar-NIRCam	NIRCam Imaging	(1) Star-Forming-Pillar

ABSTRACT

We propose to observe a starforming pillar in narrow and broadband filters with NIRCam and MIRI, resolving protostellar outflows emerging from the molecular cloud. This is a program aimed at obtaining high-quality JWST images for outreach purposes.

OBSERVING DESCRIPTION

JWST Proposal 12591 (Created: Thursday, May 7, 2026, 5:00:10PM Eastern Standard Time) - Overview

We propose to observe a starforming pillar in narrow and broadband filters with NIRCam and MIRI, resolving protostellar outflows emerging from the molecular cloud. This proposal will use NIRCam imaging mode to obtain broad and narrow-band imaging of with both NIRCam and MIRI. The entire pillar and base can be fit on a single NIRCam pointing with a FULLBOX 5TIGHT dither pattern in 6 filters: F090W, F277W, F187N, F335M, F212N, and F470N+F444W. A larger 3×3 MIRI mosaic is required to reliably fits the entire pillar using the F770W, F1130W, and F1500W filters. The images must be sufficiently deep to allow for the creation of low-noise color-composite images. The dither patterns are optimized for uniform depth, coverage, and efficient removal of cosmic rays and other artifacts. These criteria will also produce very high-quality, science-ready data products for use by the community. This proposal does not include spectroscopic observations.

Proposal 12591 - Targets - MIRI and NIRCам imaging of star forming pillar

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Star-Forming-Pillar	RA: 10 45 59.9151 (161.4996463d) Dec: -60 06 36.90 (-60.11025d) Equinox: J2000	Proper Motion RA: -6.55 mas/yr Proper Motion Dec: 2.17 mas/yr Parallax: 3.8E-4" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[H II regions, Molecular clouds, Pre-main sequence stars, Protostars]					

Proposal 12591 - Observation 1 - MIRI and NIRCcam imaging of star forming pillar

Observation	Proposal 12591, Observation 1: Pillar-MIRI										Thu May 07 22:00:10 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	Star-Forming-Pillar	RA: 10 45 59.9151 (161.4996463d) Dec: -60 06 36.90 (-60.11025d) Equinox: J2000			Proper Motion RA: -6.55 mas/yr Proper Motion Dec: 2.17 mas/yr Parallax: 3.8E-4" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[H II regions, Molecular clouds, Pre-main sequence stars, Protostars]										
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)	Column shift (deg)		Tile Order	
	3	3	10.0		10.0		0.0	0.0		DEFAULT	
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	6	6						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	20	1	1	Dither 1	6	6	333.005	
	2	F1130W	FASTR1	20	1	1	Dither 1	6	6	333.005	
	3	F1500W	FASTR1	20	1	1	Dither 1	6	6	333.005	

Proposal 12591 - Observation 1 - MIRI and NIRCам imaging of star forming pillar

Special Requirements	Before Date 01-JUL-2026:00:00:00 Group Visits within 53.0 Days Aperture PA Range 54.83544897 to 154.83544897 Degrees (V3 50.0 to 150.0) Visits Same PA
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Proposal 12591 - Observation 2 - MIRI and NIRCam imaging of star forming pillar

Observation	Proposal 12591, Observation 2: Pillar-NIRCam									Thu May 07 22:00:10 GMT 2026
	Diagnostic Status: Warning									
Observing Template: NIRCam Imaging										
Diagnostics	(Pillar-NIRCam (Obs 2)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.									
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	Star-Forming-Pillar	RA: 10 45 59.9151 (161.4996463d) Dec: -60 06 36.90 (-60.11025d) Equinox: J2000			Proper Motion RA: -6.55 mas/yr Proper Motion Dec: 2.17 mas/yr Parallax: 3.8E-4" Epoch of Position: 2000				
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=ISM Description=[H II regions, Molecular clouds, Pre-main sequence stars, Protostars]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module gap (large extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		5TIGHT		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F187N	F335M	SHALLOW4	7	1	5	5	1825.251	
	2	F212N	F470N+F444W	SHALLOW4	7	1	5	5	1825.251	
	3	F090W	F277W	SHALLOW2	5	1	5	5	1181.045	
Special Requirements	Before Date 01-JUL-2026:00:00:00									
	Aperture PA Range 59.92542306 to 89.92542306 Degrees (V3 60.0 to 90.0) Aperture PA Range 249.92542306 to 260.92542306 Degrees (V3 250.0 to 261.0) Aperture PA Range 292.92542306 to 299.92542306 Degrees (V3 293.0 to 300.0)									

Proposal 12591 - Observation 3 - MIRI and NIRCcam imaging of star forming pillar

Observation	Proposal 12591, Observation 3: Pillar-NIRCam									Thu May 07 22:00:10 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Imaging									
Diagnostics	(Pillar-NIRCam (Obs 3)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	Star-Forming-Pillar	RA: 10 45 59.9151 (161.4996463d) Dec: -60 06 36.90 (-60.11025d) Equinox: J2000			Proper Motion RA: -6.55 mas/yr Proper Motion Dec: 2.17 mas/yr Parallax: 3.8E-4" Epoch of Position: 2000				
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=ISM Description=[H II regions, Molecular clouds, Pre-main sequence stars, Protostars]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module gap (large extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		5TIGHT		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F187N	F335M	SHALLOW4	7	1	5	5	1825.251	
	2	F212N	F470N+F444W	SHALLOW4	7	1	5	5	1825.251	
	3	F090W	F277W	SHALLOW2	5	1	5	5	1181.045	
Special Requirements	Before Date 01-JUL-2026:00:00:00 Aperture PA Range 59.92542306 to 89.92542306 Degrees (V3 60.0 to 90.0) Aperture PA Range 249.92542306 to 260.92542306 Degrees (V3 250.0 to 261.0) Aperture PA Range 292.92542306 to 299.92542306 Degrees (V3 293.0 to 300.0)									